L&T ASSIGNMENT REPORT • SUBMISSION ITEMS 1 & 2

TIMER-BASED LED FLASHER PCB

PCB Design, Footprint Mapping, Layout Routing & Gerber Generation using KiCad EDA

1. Student & Assignment Identification

Items 1 & 2

1. Name of the Student	**Yuvang Chikara** | Roll / Employee ID: **2630btech1313**
Course / Section & Institution	ECE Sec: C | KIET University
2. Assignment Title	**Timer-Based LED Flasher PCB (Assignment Question 6)**
Design Software	KiCad EDA 10.0.0 (Schematic Editor, Footprint Assignment, PCB Editor, 3D Viewer, Gerber Plotter)
Core Circuit Components	NE555 Timer IC, 5 mm Red LED, Timing & Limiting Resistors (1 kΩ, 47 kΩ, 470 Ω), Capacitors (10 µF, 10 nF), 9 V Power Connector
Graded Tasks Completed	Task 1: Schematic Creation (2 Marks) • Task 2: Footprint Assignment (2 Marks) • Task 3: PCB Layout & Routing (2 Marks)

Scenario & Abstract

Engineering Scenario: A signage company requires a reliable, low-cost electronic module that periodically flashes an high-visibility LED to attract attention to public information displays and industrial warning signs. Operating as a PCB Design Engineer, this project documents the end-to-end hardware realization of a **Timer-Based LED Flasher PCB** using KiCad EDA.

Design Overview: The circuit employs an industry-standard **NE555 monolithic timing IC** configured in astable multivibrator mode. Powered from a 9 V DC supply via a 2-pin vertical header (J1), the timer continuously charges and discharges an electrolytic timing capacitor (C1 = 10 µF) through precision axial resistors (R1 = 1 kΩ, R2 = 47 kΩ) to generate a continuous square-wave output at Pin 3. This output drives a 5 mm red indicator LED (D1) through a 470 Ω series current-limiting resistor (R3).

Project Objectives & Deliverables Checklist

- **Task 1 (2 Marks):** Capture a clean, ERC-verified schematic in KiCad using standard IEEE/IEC symbols and power flags.
- **Task 2 (2 Marks):** Assign practical through-hole (THT) footprints to all 8 components for easy hand assembly and soldering.
- **Task 3 (2 Marks):** Place components within a compact rectangular board outline, route copper traces with zero DRC errors, inspect in 2D/3D views, and generate manufacturing BOM and Gerber fabrication files.

Theory, Working Principle & Design Calculations

An astable multivibrator is a free-running oscillator that alternates automatically between two quasi-stable states without any external trigger pulse. Inside the NE555 IC, three matched 5 kΩ internal resistors establish comparator reference thresholds at **1/3 V_{CC}** (3.0 V) and **2/3 V_{CC}** (6.0 V) when powered by a 9 V supply.

Charging Cycle (Output HIGH — LED ON): Initially, timing capacitor C1 (10 µF) charges from 1/3 V_{CC} toward V_{CC} through series resistors R1 (1 kΩ) and R2 (47 kΩ). During this interval, the internal discharge transistor at Pin 7 is off, and Pin 3 (OUT) is held HIGH (~7.5 V), forward-biasing LED D1 through resistor R3 (470 Ω).

Discharging Cycle (Output LOW — LED OFF): Once the voltage across C1 reaches 2/3 V_{CC} (Pin 6 Threshold), the upper comparator trips the internal flip-flop. Pin 3 switches LOW (0 V), turning the LED off, while Pin 7 (DISCH) turns on and discharges C1 solely through resistor R2 until the capacitor voltage drops back to 1/3 V_{CC} (Pin 2 Trigger), repeating the flash cycle indefinitely. Capacitor C2 (10 nF) decouples Pin 5 (CTRL) to ground to prevent false noise triggering.

OSCILLATION FREQUENCY (F)
$f = 1.44 / ((R1 + 2 \times R2) \times C1)$
$f = 1.44 / ((1\ k\Omega + 2 \times 47\ k\Omega) \times 10\ \mu F) = 1.44 / 0.95 =$
1.52 HZ

ON TIME (T_H) & OFF TIME (T_L)
$t_H = 0.693 \times (R1 + R2) \times C1 = 0.693 \times 48\ k\Omega \times 10\ \mu F \approx$
0.333 S

$t_L = 0.693 \times R2 \times C1 = 0.693 \times 47\ k\Omega \times 10\ \mu F \approx$
0.326 S

DUTY CYCLE (D)
$D = t_H / (t_H + t_L) = (R1 + R2) / (R1 + 2 \times R2)$
$D = 48\ k\Omega / 95\ k\Omega \times 100\% \approx$
50.5% (~51% SYMMETRICAL FLASH)

LED FORWARD CURRENT (I_{LED})
$I_{LED} = (V_{OUT} - V_F) / R3$
$I_{LED} \approx (9\ V - 1.5\ V - 2.0\ V) / 470\ \Omega = 5.5\ V / 470\ \Omega \approx$
11.7 MA (≈ 12 MA)

Step-by-Step KiCad Design Workflow

1. **Step 1 — Schematic Capture (Task 1):** Placed symbols for `NE555P` (U1), axial resistors (R1, R2, R3), polarized capacitor (C1), ceramic disc capacitor (C2), red LED (D1), and 2-pin connector (J1). Added `+9V`, `GND`, and `PWR_FLAG` symbols and passed Electrical Rules Check (ERC = 0 errors).
2. **Step 2 — Footprint Assignment (Task 2):** Linked each symbol to standard THT footprints (`DIP-8_W7.62mm`, `R_Axial_DIN0207`, `CP_Radial_D5.0mm`, `C_Disc_D5.0mm`, `LED_D5.0mm`, `PinHeader_1x02`).
3. **Step 3 — PCB Placement, Routing & Fabrication (Task 3):** Imported netlist into PCB Editor, defined a 40 mm × 32 mm `Edge.Cuts` boundary, routed 0.5 mm signal tracks and 0.8 mm power tracks on `B.Cu`, verified DRC = 0 errors, inspected in 3D Viewer, and plotted Gerber/Drill files.

> **Design Verification Note:** Always verify electrolytic capacitor C1 polarity, LED cathode flat-edge orientation, and NE555 Pin 1 notch alignment against component datasheets before soldering or applying 9 V power.

3. KiCad Schematic Diagram

Task 1 (2 Marks) • Submission Item 3

The schematic documents all circuit symbols, component designators, calculated part values, power flags (PWR_FLAG), and electrical net interconnections created in KiCad Eeschema (v10.0.0).

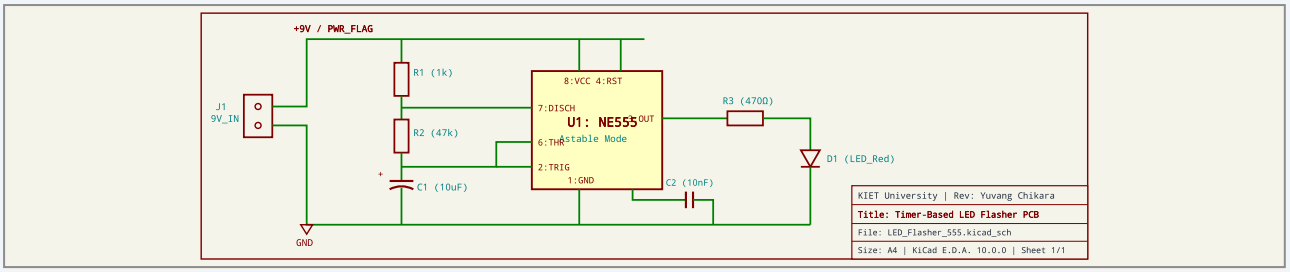


Figure 1. KiCad Schematic Editor screenshot showing NE555 astable connections, power flags, and title block.

Schematic Netlist & Pin-to-Pin Electrical Connections

Net Name	Connected Component Pins	Electrical Function
+9V	J1 Pin 1, U1 Pin 8 (VCC), U1 Pin 4 (RESET), R1 Pin 1	+9 V DC Supply & Active-High Reset
GND	J1 Pin 2, U1 Pin 1 (GND), C1(−), C2 Pin 2, D1 Cathode	Common Ground Reference (0 V)
N_DISCH	R1 Pin 2, R2 Pin 1, U1 Pin 7 (DISCH)	Open-Collector Discharge Node
N_TIMING	R2 Pin 2, U1 Pin 6 (THRES), U1 Pin 2 (TRIG), C1(+)	RC Timing & Comparator Sense Node
N_CTRL	U1 Pin 5 (CTRL), C2 Pin 1 (10 nF)	Internal 2/3 V_{CC} Reference Bypass
N_OUT	U1 Pin 3 (OUT), R3 Pin 1 (470 Ω)	1.5 Hz Square-Wave Timer Output
N_LED	R3 Pin 2 (470 Ω), D1 Anode (5 mm Red LED)	Current-Limited (~12 mA) LED Drive

Task 2: KiCad Footprint Assignment

Task 2 (2 Marks)

Using KiCad's *Assign Footprints* tool, each schematic symbol was mapped to a standard through-hole (THT) physical package with verified pad pitch and drill diameter.

Ref	Value	Assigned KiCad Library : Footprint Name
U1	NE555	`Package_DIP:DIP-8_W7.62mm`
R1, R2, R3	1k, 47k, 470Ω	`Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal`
C1	10 µF	`Capacitor_THT:CP_Radial_D5.0mm_P2.00mm`
C2	10 nF	`Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm`
D1	LED Red	`LED_THT:LED_D5.0mm`
J1	9V Power	`Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical`

4. Screenshot of Footprint After Routing Is Completed

Submission Item 4

The view below shows the close-up component footprints, numbered through-hole pads, courtyard boundaries, and completed copper tracks in the KiCad PCB Editor with zero unrouted nets (`DRC = 0 errors`).

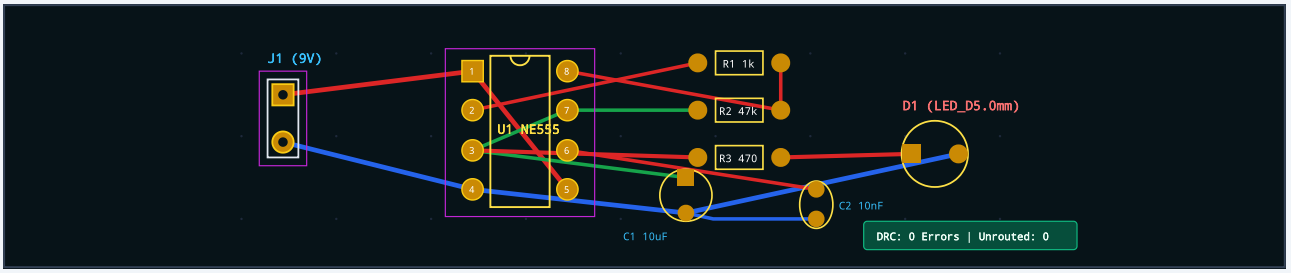


Figure 2. KiCad PCB Editor close-up showing footprints, numbered THT pads, silkscreen outlines, and completed copper routing.

5. 2D PCB View Screenshot

Task 3 (2 Marks) • Submission Item 5

The full 2D PCB layout view illustrates the complete 40 mm × 32 mm rectangular board outline (`Edge.Cuts`), corner M3 mounting holes, top silkscreen (`F.SilkS`), copper ground pour, and routed tracks.

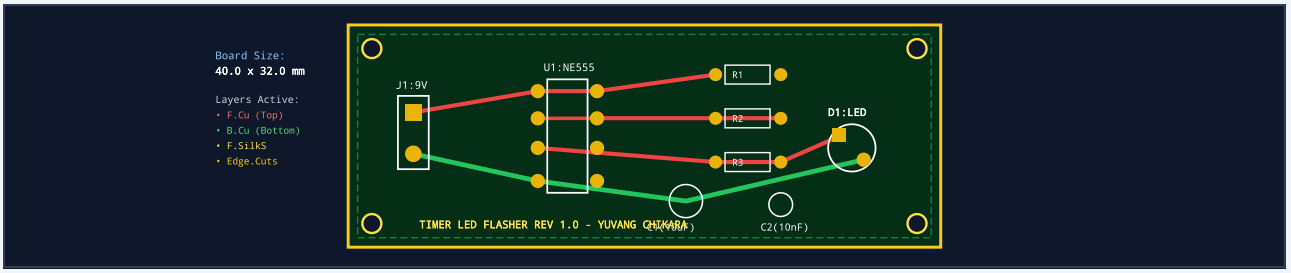


Figure 3. Complete 2D PCB Editor layout showing board outline, mounting holes, silkscreen labels, and copper tracks.

6. 3D PCB View Screenshot

Submission Item 6

The interactive 3D Viewer (`View → 3D Viewer`) validates mechanical clearances, component body heights, resistor color bands, DIP-8 IC seating, and unobstructed visibility of the flashing 5 mm red LED.

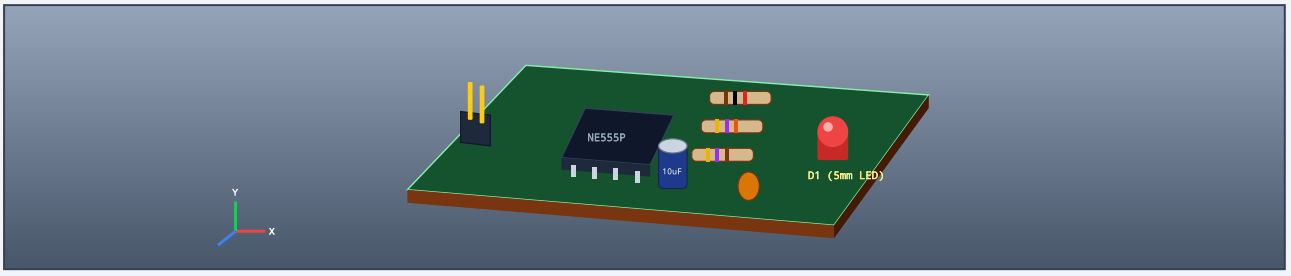


Figure 4. KiCad 3D Viewer rendering showing populated THT components, DIP-8 timer IC, electrolytic capacitor, and 5 mm red LED.

7. Bill of Materials (BOM)

Submission Item 7

Complete manufacturing Bill of Materials exported from the verified KiCad schematic (Total: 8 line items / 8 components).

Item	Ref	Qty	Value	KiCad Footprint	Component Description
1	**U1**	1	NE555	`DIP-8_W7.62mm`	Precision 555 Timer IC (Astable)
2	**R1**	1	1 kΩ	`R_Axial_DIN0207_P10.16mm`	Charge Timing Resistor (1/4 W, 5%)
3	**R2**	1	47 kΩ	`R_Axial_DIN0207_P10.16mm`	Discharge Timing Resistor (1/4 W)
4	**R3**	1	470 Ω	`R_Axial_DIN0207_P10.16mm`	LED Current-Limiting Resistor
5	**C1**	1	10 µF	`CP_Radial_D5.0mm_P2.00mm`	Electrolytic Timing Cap (25 V)
6	**C2**	1	10 nF	`C_Disc_D5.0mm_P5.00mm`	Ceramic Control-Pin Bypass Cap
7	**D1**	1	LED Red	`LED_D5.0mm`	5 mm High-Brightness Red LED
8	**J1**	1	Conn_01x02	`PinHeader_1x02_P2.54mm`	9 V DC Power Input Header

8. Gerber & Drill Fabrication Files Summary

Submission Item 8

Generated via `File → Fabrication Outputs → Gerbers (.gbr)` in RS-274X format alongside Excellon drill files (`.drl`) and verified in the KiCad GerbView utility prior to ZIP archive packaging (`LED_Flasher_Gerbers.zip`).

Generated Gerber / Drill Filename	KiCad Layer	Fabrication Purpose
`LED_Flasher_555-F_Cu.gbr`	`F.Cu`	Top Copper Layer Traces & THT Pads
`LED_Flasher_555-B_Cu.gbr`	`B.Cu`	Bottom Copper Tracks & GND Plane
`LED_Flasher_555-F_Silkscreen.gbr`	`F.SilkS`	Top Component Designators & Polarity Marks
`LED_Flasher_555-F_Mask.gbr / B_Mask.gbr`	`F.Mask / B.Mask`	Top & Bottom Green Solder Mask Openings
`LED_Flasher_555-Edge_Cuts.gbr`	`Edge.Cuts`	40 mm × 32 mm CNC Board Perimeter Outline
`LED_Flasher_555-PTH.drl / NPTH.drl`	`Excellon Drill`	Plated Through-Holes & M3 Mounting Holes